

CSC 480 – Artificial Intelligence 1
2025-26 – Spring Term
Instructor: Kenny Davila Castellanos

Homework #2: Logic and Probabilistic Reasoning

Objective

The first goal of this homework is to practice Propositional and First Order Logic on simple problems. The second goal is to practice how to represent logical statements using code. The third Goal is to use Bayesian Networks to compute joint probabilities.

Part 1. Propositional Logic (Knights and Knaves)

Solve the following problems and write out your reasoning using propositional logic. For each step, record which facts you are operating on, and which rule you are using. See the slides from class entitled, “Example: The Island of Knights & Knaves”, for an example of how to formulate your answer.

Q1-1: On the Island of Knights and Knaves, you meet two people. A says, “B is a knave.” B says, “Neither A nor I are knaves.” Who is what?

Q1-2: On the Island of Knights and Knaves, you meet two people. A says, “We are the same kind.” B says, “We are of different kinds.” Who is what?

Part 2. Propositional Logic (Proof by Resolution Refutation)

Suppose we have a knowledge base KB consisting of the following facts and rules:

- KB1: Either Heather attended the meeting or Heather was not invited.
- KB2: If the boss wanted Heather at the meeting, then she was invited.
- KB3: Heather did not attend the meeting.
- KB4: If the boss did not want Heather there, and the boss did not invite her there, then she is going to be fired.

Using the following literals:

- WANT: the boss wants (or wanted) Heather to attend the meeting
- INVITE: the boss invites (or invited) Heather to the meeting
- FIRE: Heather will be fired
- ATTEND: Heather attends (or attended) the meeting

Q2: Use resolution-refutation to prove that Heather is going to be fired (i.e., FIRE). *For each step, record which facts you are operating on and which rule you are using.* See the slide in the lecture note on Logic to show the steps in the proof. You must use the **algorithmic solution** (step by step), remember **the computer cannot take intuition shortcuts!**

Part 3. First Order Predicate Logic (CNF)

Using the predicates listed below:

- Teacher(x): x is a teacher.
- Student(x): x is a student.
- Test(x): x is a test.
- Passed(x, y): x passed y.

Q3-1: Represent each of the following sentences into a first order logic (and quantified) statement.

- 1) "Some students are also teachers".
- 2) "Every student has failed at least one test during their studies"
- 3) "There is exactly one test that every student has passed".

Q3-2: Translate each of the previous first order logic statements to a Conjunctive Normal Form (CNF). Show **every step** of the CNF conversion in detail. Be sure to incorporate **Skolemization** whenever needed.

Part 4. First Order Logic (Proof by Resolution Refutation)

Consider the following statements about creatures in a Video Game:

1. Each creature must be exactly one of the following types: Ogre, Dwarf, or Goblin.
2. Some creatures are monsters.
3. Monsters can eat some of the other creatures.
4. Creatures can defeat any creatures they eat.
5. Defeating is transitive. If x defeats creature y, and y defeats z, then x defeat z.
6. If a creature is an Ogre, then that creature can eat dwarves.
7. If a creature is a Dwarf, then that creature can defeat goblins.
8. All goblins are monsters.

Q4-1. Represent these facts and rules as sentences in first-order predicate (and quantified) statements.

Q4-2. Translate your sentences into CNFs. Show your work. Be sure to incorporate Skolemization whenever needed.

Q4-3. Use **resolution-refutation** to prove that in general, ogres can defeat goblins.

Note that, in addition to the facts explicitly stated above, you may need to write other statements to do the proof. In particular, the usage of Skolemization to create instances of certain creatures will be really helpful for your proof.

Q4-4. What other creatures can an Ogre eat? Show your work in detail.

Part 5. Liar Puzzle

Someone crashed the server, and accusations are flying. For this problem, we will encode the evidence in first-order logic formulas to find out who crashed the server. You've narrowed it down to four suspects: John, Susan, Mark, and Nicole. You have the following information:

1. Mark says: "It wasn't me!"
2. John says: "It was Nicole!"
3. Nicole says: "No, it was Susan!"
4. Susan says: "Nicole's a liar."
5. You know that exactly one person is telling the truth.
6. You also know exactly one person crashed the server.

Who did it? First, convert this information into a knowledge base in first order logic. Then, using any of the proofing algorithms studied in class, determine:

Q5-1 Who is telling the truth?

Q5-2 Who crashed the server?

Part 6. Bayesian Networks (on Paper)

Mauricio is an employee of some company who loves eating pizza, but he mainly eats pizza when working late and wants to have a late dinner. He has a job that forces him to stay late quite often, and the likelihood of having late dinner is much higher when he has to work until late. In addition, Mauricio also cares about the calorie count of the food eaten during the day, and since he also loves having burgers (mainly for lunch), he generally avoids having pizza for dinner if he already had burger for lunch that day. The likelihood of having a burger lunch is in turn affected by how much he is craving to eat bread and/or veggies. He tends to crave eating bread more often than veggies, but these are generally independent of each other. Consider the following events:

- 1) V – Wanting to eat veggies for lunch
- 2) B – Wanting to eat bread for lunch
- 3) G – Eating a burger for lunch
- 4) W – Working until late
- 5) D – Having a late dinner
- 6) Z – Having pizza for dinner

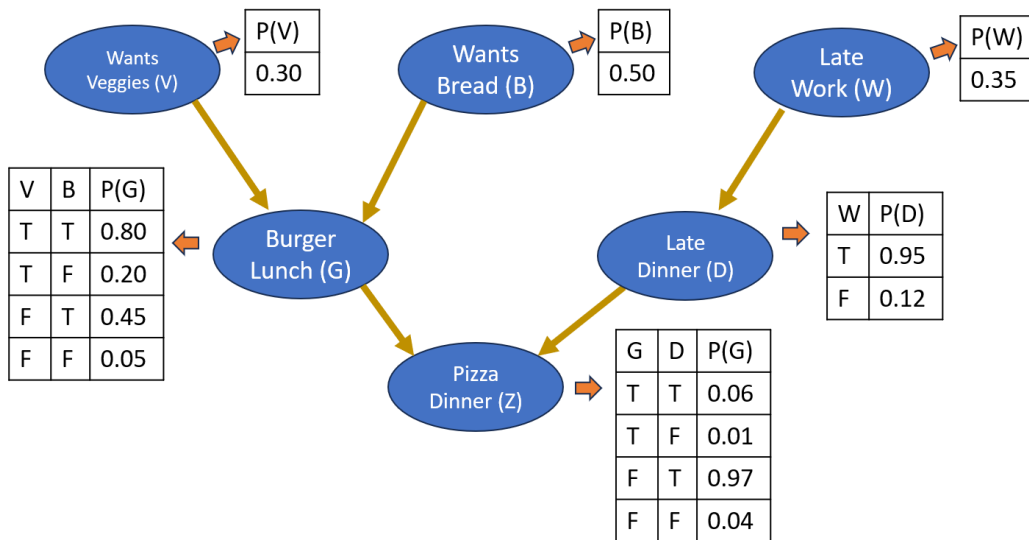


Figure 1. The Food Decision Bayesian Network

The statistical dependencies between each of these events is modeled by the Bayesian network in Figure 1. Use this information to compute the following **joint** probabilities:

Q6-1. Probability of having Pizza for Dinner on a day that Mauricio wanted veggies for lunch but not bread and yet he ate a Burger, and he is staying late for work and is having late dinner.

Q6-2. Probability of having Pizza for Dinner on a day that Mauricio wanted bread for lunch but not veggies and yet he ate a Burger, and he is staying late for work and is having late dinner.

Q6-3. Probability of **NOT** having Pizza for Dinner on a day that Mauricio wanted both bread and veggies for lunch but did **NOT** eat Burger. He is **NOT** staying late for work, but he is still having late dinner.

Part 7. Bayesian Networks in Python

A demo of how to implement Bayesian Networks using python is included in the file: **demo.py**. This code implements one of the examples studied in class (Earthquake, Burglary, Alarm, John Calls, Mary Calls). Now, use this reference to implement the Bayesian Network shown in **Part 6** of this assignment. Beyond simple queries for one specific combination of variables in the joint probability distribution, the engine allows to run more advanced queries that might require marginalization over multiple values for the unobserved variables. Use these capabilities to run the following queries, and make sure to include the results in the written report:

- Q7-1.** Probability that Mauricio has Pizza for dinner **given** that he had Burger for lunch.
- Q7-2.** Probability that Mauricio has a Burger for lunch **given** that he has Pizza for dinner.
- Q7-3.** Probability that Mauricio Works late **given** that he does **NOT** have Pizza for dinner.
- Q7-4.** Probability that Mauricio wants Veggies for lunch **given** that he has a Late dinner.

The Report

You have to prepare a written report with the results for each part of this assignment. It must have the following sections:

- 1. Introduction**
 - a. Briefly explain the whole assignment in your own words.
- 2. Solution for Part 1.**
- 3. Solution for Part 2.**
- 4. Solution for Part 3.**
- 5. Solution for Part 4.**
- 6. Solution for Part 5.**
- 7. Solution for Part 6.**
- 8. Outputs for Part 7.**
- 9. Difficulties (Optional).**
 - a. Provide a brief summary of difficulties that you faced while implementing this project.
- 10. New application (0.25 – 0.5 pages).**
 - a. Propose an entirely different application where you would use these Logic Inference Algorithms.
 - b. Describe the application in detail.
 - c. Be specific, what is the **what type of relations would you consider?**
 - d. Be concise. Do not use fillers to reach the minimum length.
 - e. Be creative. Do not use chat-GPT, it will be obvious if you do.
- 11. Feedback (0.25 – 0.5 pages)**
 - a. What did you think about this assignment?
- 12. Conclusions.**

Delivery Instructions

Submit your report in a single PDF file.

File names. The PDF file that you submit should use “[Last Name(s)], [Given Name(s)].pdf” as it appears in D2L. For example, “Kenny Davila Castellanos” (Davila Castellanos is two last names), would have to submit the homework with the name “Davila Castellanos, Kenny. pdf”. Another student named “Kenny Mauricio Davila” (Mauricio is a middle name), would have to submit the homework as “Davila, Kenny Mauricio. pdf”.

Policies

1. This homework is meant to be worked **individually**.
2. All general policies about Plagiarism and Cheating apply to this Assignment. If you plagiarize or receive code from other people, you will be caught and you will receive a score of 0, and the academic integrity violation will be filed.
3. You must not use Chat-GPT or any other code generators for completing this assignment. Doing so will be considered a form of cheating and are also eligible to receive a score of 0.
4. Do not post your solutions online and do not share them with anyone. It is your responsibility to safeguard your private data.
5. You must follow the delivery instructions.
6. Late submissions without justification might receive penalties proportional to the number of hours late. See the syllabus for concrete details on late penalties.
7. The code and documentation highlight using specific approaches for certain things. **You are expected to follow these approaches. Any different approach that you use without consulting me first is likely to receive a score of 0.**
8. Do ask for help if anything is unclear, but do it in a timely manner (e.g., by e-mail or during the Office Hours).